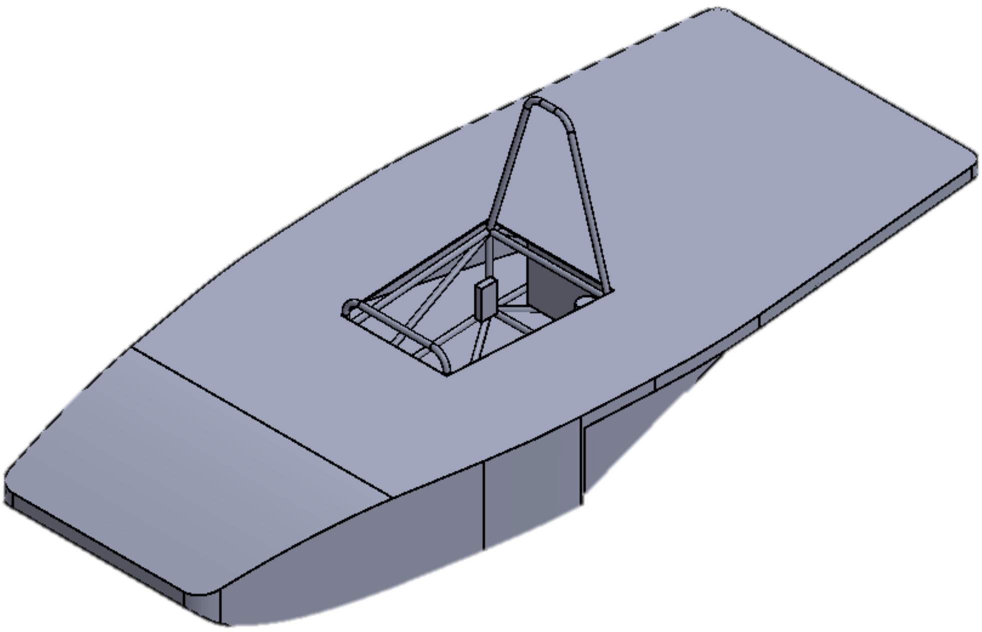
Title: - SEVC Virtual Report

TEAM STES HYPERION

SINHGAD TECHNICAL EDUCATION SOCIETY

Solar Electric Vehicle Championship

Vehicle Category: - Hyper Electric Vehicle
Team ID: - 26013



SUSPENSION DESIGN REPORT

Introduction

The suspension system connects a vehicle's chassis to its wheels and allows controlled movement while absorbing road shocks. It consists of springs, dampers, linkages, and tires, all of which work together to improve ride comfort, handling, and safety. Maintaining continuous tire contact with the road is essential for effective cushioning and vehicle control. Proper weight distribution through the suspension system also helps in reducing tire wear. Different types of suspension systems are designed to balance performance and comfort based on vehicle requirements.

Rulebook Constraints:

1) Wishbone/ A-Arm should be a seamless tube.
2) The suspension used must have a minimum total travel of 4inch (101.6 mm).

Force consideration

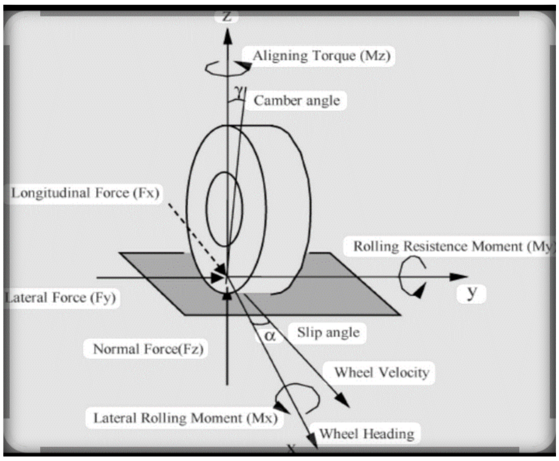


Fig No. 1.1.1 FBD of single wheel

Working Principle

Instantaneous centre is the momentary centre around which the suspension linkage pivot around. As the suspension moves, the instant centre moves due to the changes in the suspension geometry. The position of the roll centre is determined by the location of the instant centres.

Design Consideration

While designing a front suspension system, key considerations include geometry to optimize anti-dive and bump steer, ensuring minimal steering changes during suspension travel. proper damping, spring rates, and material selection are crucial for achieving a balance of ride quality and performance.

While designing rear suspension system, key considerations include anti-squat geometry.

Front Suspension System:

Suspension types	Adv.	Disadvantages.
MacPherson Strut	Straightforward design And compact design Lightweight	Limited control Lower performance
Double Wishbone	Superior Handling & stability Better Ride comfort, Improved tire contact	Heavier, costly to manufacture

Table. No. 1.1.1. Comparison

By comparing above results we selected double wishbone suspension system for front suspension

Rear suspension system:

Suspension types	Adv.	Disadv.
Trailing arm	Good stability, effective for off-road, simple and strong construction	Limited Adjustability
Swing Arm	Cost effective Good ride comfort	Less Responsive Handling, Increased

Table No. 1.1.2. Comparison

By comparing above results we selected swing arm suspension system for rear suspension

Material Selection:

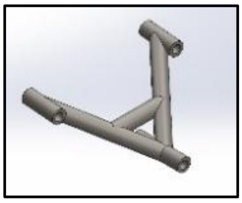
Pugh's method was chosen for material selection because it quantitatively ranks multiple options.

S R. N O	SELECTION CRITERIA	AISI 4130	AISI 1018	AL 70 75
1	Strength	+	=	-
2	Weight	=	=	+
3	Weldability	+	+	-
4	Cost	=	+	-
5	Machinability	+	+	+
6	Availability	=	+	+
	TOTAL	3	4	0

Table No. 1.1.3 Pugh's Matrix

Based on the Pugh Matrix analysis, AISI 1018 is recommended for critical structural components.

Component Design



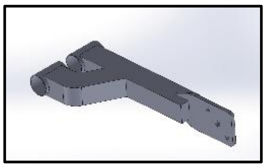


Fig No. 1.1.2 wishbone **Fig No. 1.1.3 trailing arm**

Front Suspension Design- (Double wishbone damper to lower wishbone)

CAD Analysis: -

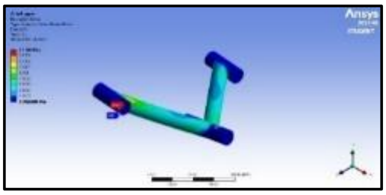


Fig No. 1.1.4 Wishbone Stress analysis

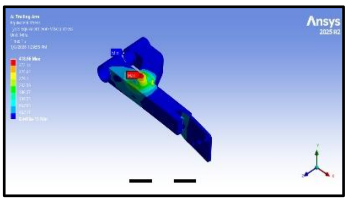


Fig. No. 1.1.5 Trailing arm stress analysis

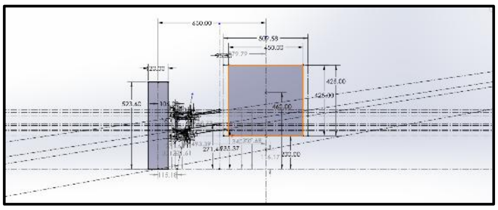


Fig No. 1.1.6 Front view geometry

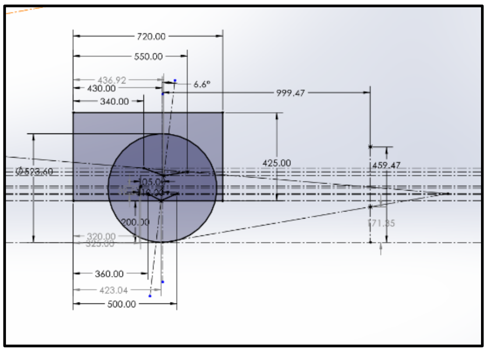


Fig No 1.1.7 Side view geometry

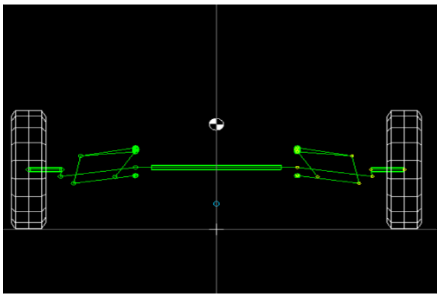


Fig No. 1.1.8 Front suspension simulations in lotus shark

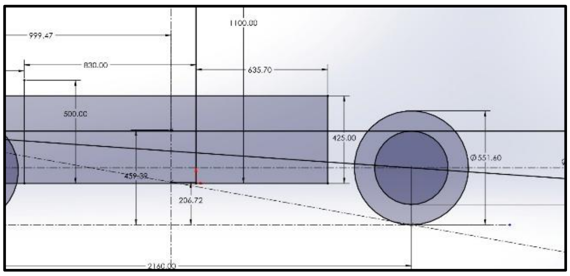


Fig No. 1.1.9 Side View Swing arm geometry

Features/ Aspect	Swing arm mono shock suspension	Swing arm dual suspension
Stability	Less stable	Enhanced stability and control

Performance under load	Good, but can be harsher	Improved ride quality and comfort
Off-Road Capability	Limited performance in rough Terrain	Excellent performance on uneven surfaces
Shock Absorption	Limited dampening control	Superior shock absorption and performance
Weight	Generally lighter	Heavier due to additional components
Design Complexity	Simpler design with fewer components	More complex due to dual shocks

Table No. 1.1.4 Rear Suspension Comparison

Based on above comparison we selected Swing arm with dual shock suspension

Calculations:

For the calculations of Suspension Some prerequisites are necessary.

Wheelbase (L) = 2160mm
Weight Distribution Ratio Chosen = 60:40
Weight on Front Wheels (Front)
= 283.02*0.6 = 169.8kg ≈1665.238N
Weight on Rear Wheels (FRear) = 283.02*0.4
= 113.2kg ≈ 1110.49N
Front Track Width (t) = 1300mm
CG to Roll Axis (H) = 343.83

The max longitudinal acceleration Ax = 1.2g
The max lateral acceleration Ay = 1.4g

However, the maximum acceleration experienced by the vehicle is during cornering accompanied by braking and given by: √1.22+1.42 = 1.61g

Lateral load transfer (LLT)= $\frac{Ay*h}{t}$ = $\frac{1.4*459.39}{1300}$ =0.494

Ride Rate calculation:

Front ride rate = $\frac{(LLT+0.4)*W*9.81}{wheel\ travel}$ =20.68N/mm

Rear ride rate = $\frac{(LLT+0.6)*W*9.81}{wheel\ travel}$= 25.30N/mm

Front roll rate = $\frac{K_{RF}*t_{front}{}^{2}}{2}$
= 17474600N-mm /rad

Rear roll rate = $\frac{K_{RF}}{2}$= 12.65N-mm/rad

Roll gradient = $\frac{W*9.81*H}{front\ roll\ rate+rear\ roll\ rate}$= 0.054

Roll over calculation:

$a_y = \frac{\frac{Tb}{2l}g\cos\theta+Hax\sin\theta}{H\cos\theta}$ = 10.57g/rad
longitudinal acceleration is taken as zero
θ=16.69

Roll over calculation upon cambering.

On 0.61 camber change

$\theta_n = tan^{-1}\left(\frac{\frac{T}{2}+r\sin\gamma1}{l}\right)$ = 16.62

On 0.02 camber change

$\theta_n = tan^{-1}\left(\frac{\frac{T}{2}+r\sin\gamma1}{l}\right)$ = 16.69

$h^n = \frac{\left(\frac{T}{2}+r\sin\gamma1\right)b}{l}\cos\theta n$ = 1.911

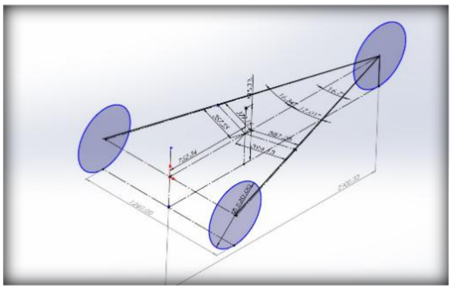


Fig No. 1.1.10 Force diagram for Tipping analysis

Tipping threshold:

The value of Fc for a three-wheeler involves the longitudinal placement of CG as well as its height and the vehicle track. the largest value of lateral acceleration (ay) that the vehicle will experience is equal to the coefficient of lateral friction at the tires. The lateral forces are simply whatever is needed to keep the vehicle from sliding.

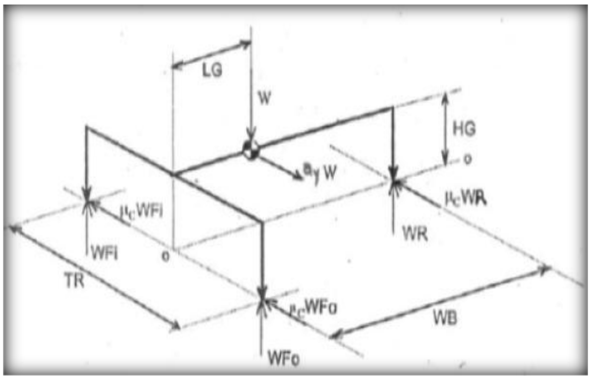


Fig No. 1.1.11 Force diagram for Tadpole vehicle

$a_y \geq \frac{TR(WB-LG)}{2(WB)(HG)} = F_c = 1.07$

Force calculation:

Total longitudinal load transfer-

ΔW_x= (W∗ HCG∗ Ax)/wheelbase = 72.23kg

Total lateral load transfer-

ΔW_y = (W* HCG* Ay)/trackwidth = 140kg

Diagonal weight transfer-

$\Delta W_D = \sqrt{W_x{}^2 + W_y{}^2} = 157.53$kg

Max weight on one wheel-

$\Delta W_M = W_S\,(front) + \Delta W_D = 229.76$kg

Max weight on one wheel-

$\Delta W_M = W_S\,(rear) + \Delta W_D = 297.83$kg

Force due to longitudinal transfer-

= 1.2*9.81*(36.11+78.765) = 1352.30N

Force on single A-Arm,

=1352.30/2 = 676.15N

Force due to lateral transfer,

= 1.4*9.81(70+78.765) = 2043.13N

Force on single A-Arm,

= 2043.13/2 = 1021.56N

Bump and Cornering:

Total load on outer wheel is calculated by adding load due to Un-sprung mass,

Gyroscopic effect and centrifugal force.

Cornering Force -

Vertical load acting on each wheel

= 642.30N

Vertical force due to centrifugal couple

$= \frac{total\ mass\ with\ driver * velocity^2 * height\ of\ CG}{2 * cornerning\ radius * track\ width}$

=73.63N

Vertical force due to gyroscopic effect= I ω ωp

=4*(mass of each wheel)*(radius of tire)* $\frac{velocity\ of\ vehicle^2}{2 * radius\ of\ tire * cornering\ radiius}$

=199.89N

Net vertical load acting on wheel

= 642.30+73.63+199.89 = 915.82N

Lateral force = Coefficient of friction * Net vertical load

= 915.82 * 0.8 = 732.66N

Bump Force-

3g force is considered for bump force

= 3*915.82 = 2747.46 N

DESIGN OF SPRING:

Design of spring (FRONT):

From the triangle, force exerted on the spring due to bump,

F Bump =2747.46 N force exerted on spring due to weight of vehicle,

F wheel = 732.66N

Therefore, Spring force,

F= F Bump-F wheel 2747.46 –732.66N

=2014.8N

Spring constant, K = F/x = 2014.8/50 N/mm ≈ 40.296N/mm

The stiffness of a Helical coiled spring is given by,

K = $G.\ d$^4 / (8*D^3) N

For front spring,

G=81370N/mm2,d=7mm,D=48mm, K=40.296N/mm,

On substituting, N= 6

Design of Springs (REAR)-

Force exerted in the spring due to bump, FBump = 3678.04N

Force exerted on spring due to weight of vehicle,

Fwheel = 1226.011N

F=FBump-Fwheel= 3678.04–1226.01N =2452.03N

Spring constant, K = F/x =2452.03/50 N/mm ≈ 49.04N/mm

The stiffness of a Helical coiled spring is given by,

K = $G.\ d$^4 /(8*D^3)N

For rear spring, G = 81370 N/mm2, d = 7 mm, D = 48 mm, K= 49.04N/mm on substituting, N= 4

SR No	Parameter	Values
1.	Wheelbase	2160mm
2.	Track width	1300mm
3.	Camber	0

4.	Caster	6.6
5.	Toe	0
6.	King pin inclination	11.6
7.	COG height	459.39mm
8.	Roll centre	116.17mm
9.	Roll camber	-0.63 to 0.26
10.	Ride height	200mm
11.	CG from front axle	999.47mm
12.	CG from rear axle	1160.53mm
13.	Sprung mass	224.62 kg
14.	Un-sprung mass	58.4kg

Table No. 1.1.5 Results and Parameters

STEERING DESIGN REPORT

Introduction

The steering system of a vehicle is a component that enables the driver to control the direction of the vehicle by turning the front wheels. It ensures that the vehicle follows the desired path by converting the rotational movement of the steering wheel into the angular movement of the wheels.

Rulebook Constraints:

The steering system must comply with the following constraints:

1. The steering wheel must be mechanically connected to the front wheels.

2. Cut-out steering wheel designs such as 'H' and '8' shapes are strictly prohibited.

3. Free play in the steering system must not exceed 7 degrees.

4. The minimum diameter of the steering wheel must be 254 mm.

5. The steering wheel position must be:

Below the driver's shoulder line, and

Above the driver's abdomen.

6. A mechanically driven rack and pinion steering system is mandatory and must include positive steering stops.

Dimension Parameters:

Wheelbase	2160mm
Front Track Width	1300mm
Ground clearance	200mm

Table No. 1.1.6. Dimension Parameters

Steering Geometry:

Parameter	Value
Steering arm length	110mm
Steering arm angle (α)	25°
Distance between front axle and rack (d)	100mm
Tie rod length	262.30mm
Turning radius	3259.14mm
Rack travel	67.16 mm
Ackermann Percentage	99.53%

Table No. 1.1.7 Geometry parameters

Design consideration:

Ackermann steering geometry is utilized, the inner wheel turns at a sharper angle than the outer wheel, allowing both wheels to follow

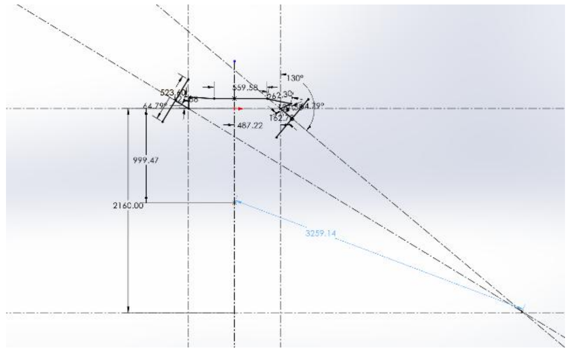


Fig. No. 1.1.12 Steering geometry

Component Selection:

Pugh's method was chosen for material selection because it quantitatively ranks multiple options.

SR. NO	SELECTION CRITERIA	EN8	EN24T	AL 7075
1	Carbon percentage	=	=	-
2	Ultimate strength	=	=	-
3	Density	=	=	+
4	Corrosion resistance	-	=	+
5	Machinability	-	+	=
	TOTAL	-2	+1	0

Table No.1.1.8 Pugh's matrix for material selection of rack and pinion

Based on the Pugh Matrix analysis. EN24 is recommended for high-strength applications.

SR.NO	SELECION CRITERIA	EN24	AISI1018	AL2024
1	Strength	+	-	=
2	Weight	-	=	+
3	Corrosion resistance	-	=	+
4	Machinability	-	+	=
5	Thermal conductivity	-	-	+
	TOTAL	-3	-1	+3

Table No. 1.1.9 Pugh's matrix for material selection of Steering Gearbox casing

Based on the above Pugh Matrix, AL 2024 is the most favorable material.

Working Principle:

We utilize a mechanically driven rack and pinion system which converts rotational movement of the pinion into linear displacement of the rack, enabling precise control and efficient operation in various applications.

CAD assembly:

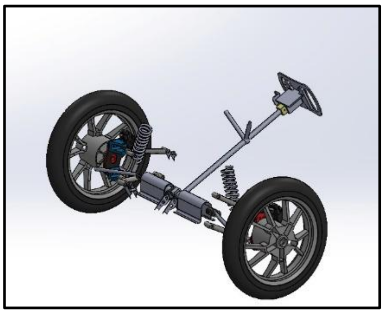


Fig No. 1.1.13 Steering mechanism assembly

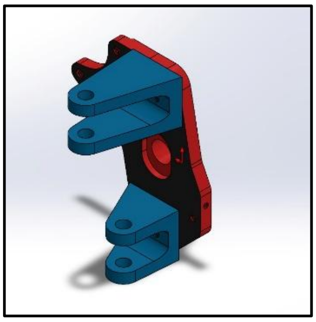


Fig No. 1.1.14 Steering Knuckle

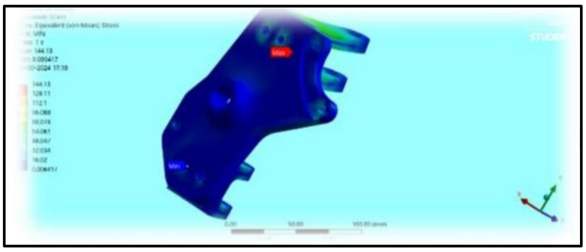


Fig No. 1.1.15 Stress analysis

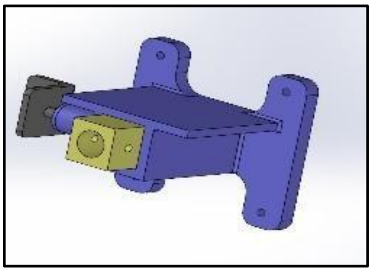


Fig No. 1.1.16 QRM

Fig No. 1.1.17 Steering

Formulation and Calculations:

Gear calculation: Min no of teeth required on pinion to avoid interference

Zp = $\frac{2ha}{m\sin^2\theta}$

Where ɵ is pressure angle 20º

Zp = $\frac{2}{\sin^2\theta} = 17.91$=18

Hence the minimum number of teeth on the pinion is 18.

Hence, we are taking it as 25.

Module Zp = Number of teeth on pinion = 25 γ =Lewis factor

Cs = service factor (assume 1 for uniform load and shock)

Np = rpm of pinion

Cv = velocity factor =

3/3+v =3/3+1 = ¾ b = face width of pinion (assume 10mm)

Sut = ultimate strength (EN24 = 850MPa)

fs = factor of safety taken as 1.5

Kw = power in kW = $\frac{Mt2\pi n}{60\times10^{6}}$

By Beam Strength Criteria,

Module=m= $\left[\frac{60\times10^{6}}{\pi}\times\{\frac{K.Cs\,(fs)}{Zp.Np.Cv.(b)(Sut)y}\}\right]$^1/3

m = 1.128≈1.5

Diameter of pinion

Module = m = 1.5

Zp = Number of teeth on pinion = 25

D= Diameter of pinion

m=$\frac{D}{ZP}$

Therefore, Diameter of pinion = 37.5mm

Steering ratio:

$\frac{rack\ travel}{2\pi r}\times\frac{360}{inner\ wheel\ angle}$

steering ratio = 4.3

PARAMETER	PINION	RACK
Module	1.5	1.5
Number of teeth	25	26
Face width	15	15
Diameter	37.5	18

Table No. 1.1.10 Rack and Pinion specification

Steering Effort:

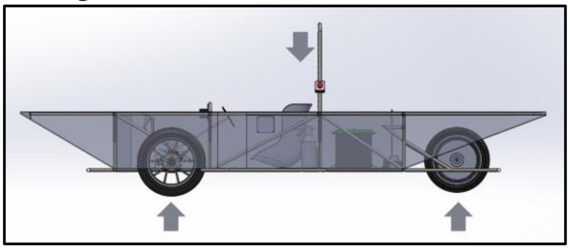


Fig No. 1.1.18 FBD of vehicle

Let, Reaction on front axle= R_A

Reaction on rear axle= R_B

Weight of the vehicle with driver=283.02kg

Taking moment at point B,

283.02×9.81×999.47= R_A×2160

R_A=1284.61N

Therefore, Force on one wheel =642.30N

Coefficient of friction of wheel = 0.8

Friction force = weight on one wheel × coefficient of friction

=513.84

Scrub radius =115.18mm

Torque on one wheel = friction force × scrub radius

= 59184.55N-mm

Force on tie rod =$\frac{Torque\ on\ one\ wheel}{steering\ arm\ length}$

=538.03N

Torque on pinion = Force on tie rod × Radius of pinion

=10088.19 N-mm

Steering effort =$\frac{Torque\ on\ pinion}{steering\ wheel\ diameter}$

=36.68N

Wear Strength Criterion:

The failure of the gear tooth due to pitting occurs when the contact stresses between two meshing teeth exceeds the surface endurance strength of the material.

$Sw=b\times Q\times dp\times K$

Q=$\frac{2\times Zg}{Zg-Zp}=\frac{2\times26}{26-25}$=52

K=$\left[\sigma c2\times\cos\emptyset\times\frac{\sin[\emptyset[\frac{1}{\epsilon1}-\frac{1}{\epsilon2}]]}{1.4}\right]$

$\sigma_c=0.27\times(BHN)kgf/mm^2$

=0.27 × (9.81) (BHN) N/mm^2

Where the BHN is Brinell Hardness Number,

Φ=20

ϵ_1=ϵ_2=205Gpafor EN24 Hence,

K= 0.1572× $\left[\frac{BHN}{100}\right]^2$

$Sw=Peff\times FOS$

The effective load is given by,

$$Peff=\frac{6\times10^{6}}{\pi}\{\frac{[kW]Cs}{mZnCv}\}$$

(By wear strength calculation)

10×45×1.26×0.1570$\{\frac{BHN}{100}\}$= 803.627 ×1.5

BHN = 367.986

To avoid pitting failure, we need to harden the rack and pinion up to 450 BHN. This can be done by any one of the hardening processes Results:

Ackermann angle	25.21˚
Max inner wheel turning angle	40˚
Scrub radius	115.18mm

Table No. 1.1.11 Parameters and Values

BRAKING SYSTEM DESIGN REPORT

INTRODUCTION

The braking system is a vital safety component that ensures vehicle stability by converting kinetic energy into heat through friction. This process allows for controlled speed reduction or complete stops, primarily categorized into hydraulic and pneumatic systems. Hydraulic systems, typical in passenger vehicles, utilize disc brakes with calipers or drum brakes with hydraulic pressure to generate friction. In contrast, pneumatic systems use compressed air to provide the high-load stopping power required for heavy commercial vehicles.

RULEBOOK CONSTRAINTS

The braking system for this solar car must feature two independent hydraulic circuits capable of simultaneously locking all wheels and holding a fully laden vehicle on a 20 degrees incline. The system relies on Pascal's Law, which dictates that pressure applied to a confined fluid is transmitted undiminished in all directions. To meet safety and structural requirements, the assembly must include a scatter shield, a brake over-travel switch, and a steel, aluminum, or titanium foot pedal capable of sustaining a force of 2000N.

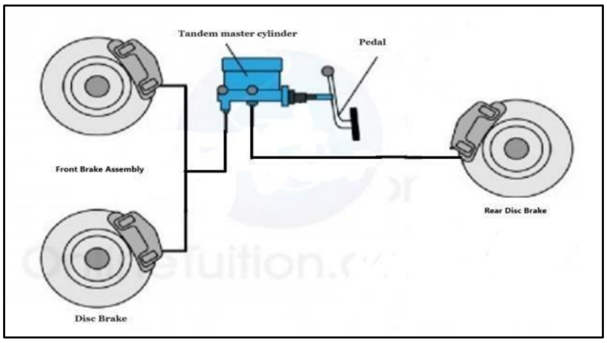


Fig 1.1.19 Braking Circuit

BRAKE PEDAL

The brake pedal serves as the primary mechanical interface in a braking system, transmitting force from the driver's foot to the master cylinder's push rod. Manufactured from mild steel, the component is chosen for its high strength, durability, and fatigue resistance, allowing it to endure repeated loads without permanent deformation. To ensure effective stopping power while maintaining driver comfort 4:1 pedal ratio is utilized to provide the necessary mechanical advantage.

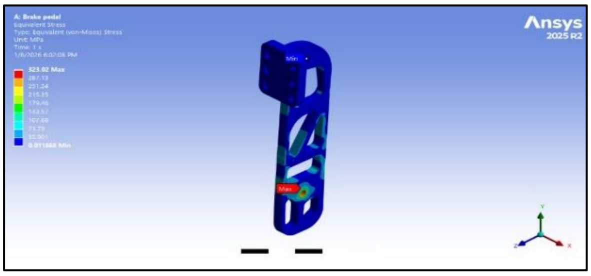


Fig 1.1.20 : Brake pedal

MASTER CYLINDER:

The master cylinder is a critical braking component that converts mechanical pedal force into hydraulic pressure. This specific vehicle utilizes two single-piston TVS Girling master cylinders with identical bore diameters to independently power the front and rear axles. This dual-circuit configuration provides precise brake force distribution and enhances safety; by keeping the hydraulic circuits separate, the vehicle retains braking capability on one axle even if the other circuit fails.

Fig No. 1.1.21 Master Cylinder

BRAKE HOSE:

The purpose of the brake line is to transfer the pressure from the master cylinder to the brake caliper. Rigid steel braided hoses and Flexible steel- braided brake hose is used in the vehicle because it is quite flexible and can transfer the pressure to the brake calipers without any distortion. These hoses conform to the SAE J 100 R7 standard and are designed with a working pressure rating of 210 bar.

Fig No 1.1.22- Rigid Hose

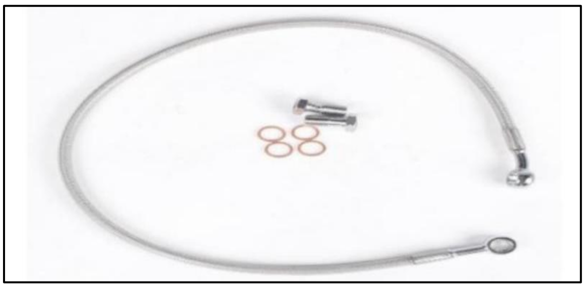


Fig No 1.1.23: Steel braided flexible hoses

BRAKE CALIPERS:

Acting as the mechanical "clamping" component of a vehicle's braking system, brake calipers convert hydraulic pressure from the master cylinder into the physical force needed to press the brake pads against the spinning rotor. These components are generally categorized into two designs: fixed and floating. Because of their ability to provide superior clamping force and more consistent pressure, fixed calipers are typically utilized for front-wheel braking, where the majority of a vehicle's stopping power is required.

Fig No 1.1.24 fixed caliper

BRAKE ROTOR

Disc brake rotor is that part of the braking system to which the brake pads are squeezed against which results in stopping of vehicle. When the brake pads squeeze against the disc brake rotor kinetic energy is converted into thermal energy and due to this conversion of energy, the vehicle decelerates and eventually stops. The material used for brake rotor SS 410.

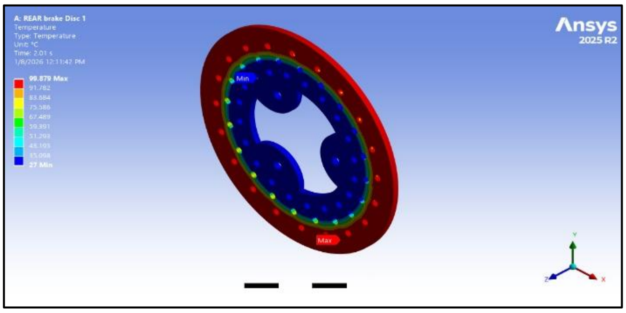


Fig No. 1.1.25 Rear Brake Disc (Thermal)

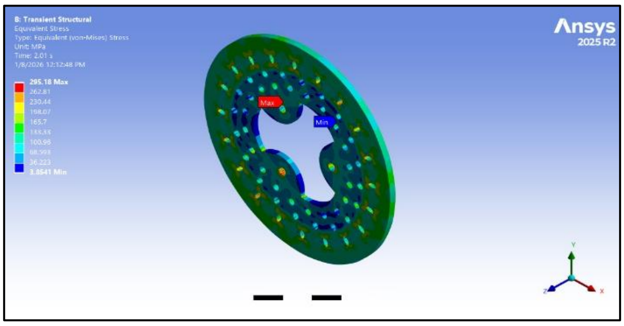


Fig No 1.1.26 Rear Disc Brake

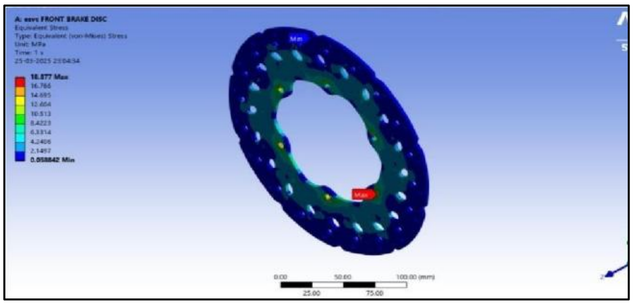


Fig No. 1.1.27 FrontBrake Disc

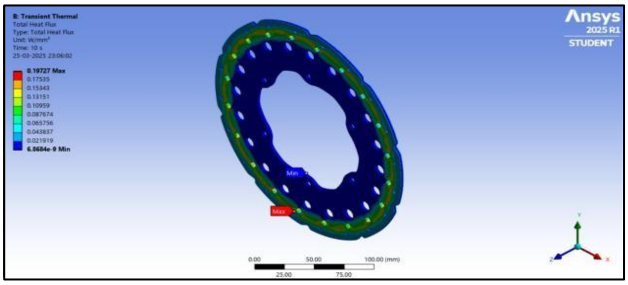


Fig No 1.1.28 Front Disc Brake (Thermal)

Braking Specifications:

TMC Bore Diameter	**19.05mm**
Caliper piston diameter (front & rear)	**32mm**
Mass of the vehicle	**283.02 Kg**
Radii of tyre	**261.8 mm**
C.G. from ground	**459.39 mm**

C.G. from front axle	**999.47 mm**
Wheelbase	**2160 mm**
Coefficient of friction between pad and disc	**0.4**
Coefficient of friction between tyre and road	**0.75**
Pedal Ratio	**4:1**

Table No. 1.1.12 Parameters and Values

BRAKE CALCULATIONS:

A Radius of Disc:

To determine the radius of the disc we compare the system generated torque and the required torque and then calculate the radius of the front and the rear disc. The applied braking torque must be greater than the required braking torque; by using this condition we calculate the radius of the disc. The optimum pedal ratio must be between 4:1 and 7:1. But to get the required clamping force for brake caliper we choose the pedal ratio as 4:1. As the main criteria for the braking is lock up the wheels to stop the car the pedal force for design consideration is taken as 250N.

The force transmitted to pushrod = Force applied on pedal × Pedal ratio

Fp = 1000 N

Where, Fp = Force applied on pushrod

Thus, pressure produced on the TMC piston

= Fp / area of TMC piston

= 1000/(A_1)

= 3508771.93 Pa

Where, A_1 = Area of TMC piston,

Fp = Force applied on pushrod

As the fluid is same throughout the system, Therefore,

pressure applied to the caliper = Fc/ Area of caliper piston

= Fc / 2(A2)

Fc = 2736.84 (front)

Where, Fc = clamping force of front caliper

A2= Area of caliper piston (front)

Static Load Calculations:

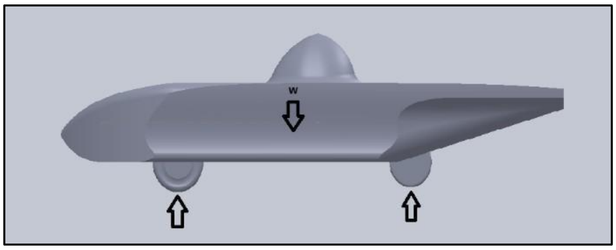


Fig 1.1.29 FBD OF vehicle

The load or force applied on the ground by the vehicle due to its weight while being at static condition is calculated. The weight of the Vehicle is divided into 2 parts, on the front axle tires (F_f) and the rear axle tires (Fr), using the following equation,

$F_f + F_r = mg$

Load on front axle,

$F_f * wb = mg * x_r$ …[X_r=CG from rear axle]

$F_f = mg * x / w_b$

$F_f = 283.02 * 9.81 * 1.16 / 2.16$

$F_f = 1491.686$ N

Similarly load on rear axle,

$F_r = 1284.74$ N

Dynamic load conditions:

Dynamic load on front axle, D_{Ff}

$D_{Ff} = F_f + (h.\mu.mg) / W_b$

$Df_f = 1934.47$ N

Dynamic load on rear axle, Dfr

Dfr = Fr - (h.µ.mg) / Wb

Dfr = 841.958 N

Dynamic weight transfer

Dynamic load on front = 1934.47 N

Dynamic load on rear = 841.958 N

Front% = 1934.47/2776.426*100

Front % = 69.67 %

Rear % = 841.958/2776.426*100

Rear % = 30.32 %

Deceleration of the vehicle

Wsin(θ)+ ma = µ(Ff + Fr)

283.02*a=0.75*(2776.426)

a= 7.357 m/s^2 or a = 0.749 g

Total braking force

Fb = m.a

Fb = 2082.18 N

Braking Force on front wheel
Dff = 1934.47 N
Fbf = μ.Dff
Fbf = 773.79 N

Braking force on rear wheel
Dfr = 841.958 N
Fbr = μ. Dfr
Fbr = 336.783 N

Braking torque on front wheel
Tf = Fbf * radii of tyre
Tf = 201.185 Nm.

Braking torque on rear wheel
Tr = Fbr * radii of tyre
Tr = 87.56 Nm

For Front Disc:
Outer Diameter of disc = 220mm
Inner diameter of disc = 170mm
Effective radius = $1/3(D^3-d^3/ D^2-d^2)$
Effective radius = 98.03mm or 0.098m
Torque = 0.098 * 2736.84
= 268.21 Nm
Hence, 268.21 > 201.185
Torque generated is greater than required torque for braking.
So, design is safe.

For rear Disc:
Outer diameter = 200mm
Inner diameter = 150mm
Effective radius = $1/3(D^3-d^3/ D^2-d^2)$
= 88.09mm or 0.088m
Torque = radii * clamping force
= 0.088 * 5473.68
= 481.68 Nm
Hence, 481.68 > 87.56
Torque generated is greater than required torque for braking. So, design is safe.

Stopping distance calculations
The deceleration of the vehicle is 7.357 or 0.749 g
The total stopping distance is the summation of the individual distance associated with two time interval

1) reaction time of the driver.
2) stopping time when deceleration is Constant.

To find the time required to stop the vehicle
V= u + at
T = 1.43 sec
The reaction time is considered to be 0.5 sec
T= 1.93
S = ut + ½ at^2
S= 7 m
So, the total stopping distance of the vehicle is 7 meters and stopping time 1.93 s

ELECTRICAL SUBSYSTEM

Motor Design Report: -
Type: - Hub Motor

Fig No. 1.1.30 Hub Motor

More Specifications: -

Rated Power	3000W
Max. Power	4000W
Rated Voltage	48V
Back EMF	44.55V
Continuous Current	50A
Max. Current	75A
Max torque	184 Nm
Rotating speed	450-900 rpm
Max efficiency	85-92%

IP Rating	IP67
Derating Temperature	>70 °C
Cutoff Temperature	120°C (cutoff in 5 - 10 seconds)
RMS	69.4A
Peak RMS	98.2A
Insulation class	Class B (130°C)
Dimension of Motor	14inch*3.5inch

Table No. 1.1.13 Motor Specifications

Selection Criteria and Calculations: -

Hub motor is an advanced motor commonly used in electric vehicles and bikes. Unlike traditional motors that use belts or chains.

A hub motor is built directly into the wheel, making it more efficient by reducing power loss. Hub motors are brushless and don't use mechanical parts like brushes to run.

This gives them many benefits, such as better efficiency, quieter operation, and a longer lifespan. They are also lighter, respond faster, and offer smoother torque and speed control, making them ideal for electric vehicles and other modern machines.

Torque Calculations: -

1)Theoretical Resistance (F-Drag): -

Drag force: -

$F_D = 1/2 * Cd * V^2 * A * \rho$

where, Cd = Coefficient of drag = 0.35

V = speed of the object relative to the fluid = 16.662

A = cross sectional area = 1 m^2

ρ = density of air = 1.225 kg/m^3

$F_D = ½ * 0.35 * 277.55 * 1 * 1.225$

$F_D = 59.5N$

Rolling Resistance: -

(Fr) = Cr*W

Value of Cr can be calculated as:

Cr=0.005 + (1/p) * [0.01+0.0095(v/100)2]

Cr = 0.0092

where, p = Tire pressure (2.41 bar)

v = Velocity of vehicle = 16 m/s

Cr = 0.0092

M = 283.02 Kg (with driver)

W = 283.02 *9.8 = 2773.596N

Fr = Cr*W= 25.5N

Acceleration Force: -

Fa = ma

Where, m = mass of vehicle (283.02Kg)

a = acceleration of vehicle (assume 0.7 m/s^2)

Fa = 283.02*0.7 = 198.114N

Initial Torque Required for Vehicle:

The minimum torque required for the motor to start initially is calculated by considering rolling resistance and acceleration force.

The tractive effort required = rolling resistance + Acceleration force

=25.5+198.114

=223.614 N

Starting torque (T_{st})=Tractive effort* radius of wheel

=223.614*0.27

= 60.37 Nm

Dynamic Condition:

Considering dynamic condition, the minimum torque required for the vehicle in running condition is calculated by considering rolling resistance and drag force.

The tractive effort required = rolling resistance + Drag force

=25.5+59.5

=85N

Running torque (Tr)=Tractive effort* radius of wheel

=85*0.27 = 22.95 Nm

Gradeability conditions:

Considering conditions where the vehicle has to start from an incline of 12°, we are considering gradeability conditions.

Fhc = w*sinα =2773.596*sin12 = 576.66 N

We consider the acceleration 0.1 m/s2 because in gradeability condition speed is very low hence the acceleration is also low.

Acceleration force F = ma

= 283.02*0.1

= 28.3 N

Total Tractive effort = Rolling resistance + Acceleration force + Grade resistance
= 25.5 + 28.3 + 576.66 = 630.4 N
Thus,

Torque required for moving vehicle on inclination
= Tractive effort * Radius of wheel = 630.4 * 0.27 = 170.2 Nm

Calculations: -

The power at which the motor operates to deliver a torque of 60.37 Nm at initial condition is, = 3.91 Kw

Current at initial conditions: -

So, from a battery of 51.2V the motor will draw a current of I = 3.91 kW/51.2V = 58.5 A

Current at Nominal conditions: -

At nominal conditions the current drawn by the motor is 3000W/51.2V= 62.5A

Current for Gradeability conditions:

The torque requirement at wheels for hill climbing is 170.2 Nm.
Power at which motor operates to deliver a torque of 170.2 Nm is, = 3.81KW
The current drawn by motor during hill climb is,
= 3810 /51.2 = 74.41 A

Vehicle speed calculations: -

Tire diameter: 551.6 mm
Tire RPM: 615 RPM is considered because running torque is 22.95Nm and the corresponding rpm in testing report of motor is 615rpm.
Circumference of tire = π x D
= π x 551.6 mm =1732.9mm = 1.73 m
Distance covered by wheel in one minute:
= 615 x 1.73 = 1063.95 meter/min
Speed of vehicle in m/s:
=1063.95/60 meter/sec = 17.73 m/s =17.73 x3600/1000 = 63.83 km/h

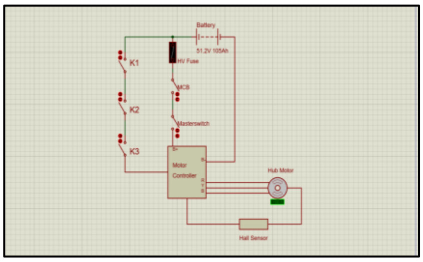


Fig. No.1.1.31 Motor Simulation Circuit Diagram

Motor controller Report: -

We have used Kelly's KLS7230N programmable motor controller for BLDC.

Specification	Details
Rated Voltage	72V
Lower Cutoff Voltage	40V
Upper Cutoff Voltage	90V
Current Rating	90A
Controller Type	Sinusoidal controller
Peak current	270A
Derating Temperature	>70°C
Cutoff temperature	100°C
Rated Power	4.3KW
Peak Power	12.9KW
Efficiency	98%
IP rating	IP66
Dimensions of controller	194mm*126 mm*71.2mm

Table No. 1.1.14 Motor Controller Specifications.

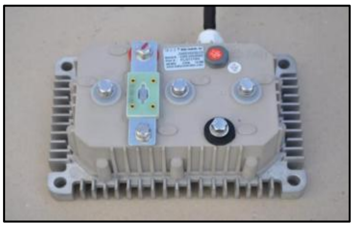


Fig. No. 1.1.32 Kelly's KLS7230N Motor

Selection Criteria: -

KLS7230N is a programmable controller.
1) It has a voltage range of 36V to 86V which is compatible with our 48V Hub motor.

2) The power of the controller is,
4.3KW

3) As the controller can be programmed, we can configure the parameters according to our needs.

4) Controller has sinusoidal commutation and has 98% efficiency.

Programmable parameters of controller: -

1) Maximum speed is set to 1000rpm as maximum rpm of motor is 900rpm.
2) Max REV speed is 30% for safety.

Parameter	Value
Max speed	1000rpm
Max FWD speed	100%
Max REV speed	30%
TPS dead High	80

TPS dead Low	20
Motor Current	100%

Table No. 1.1.15 Motor Controller Programmable Parameters

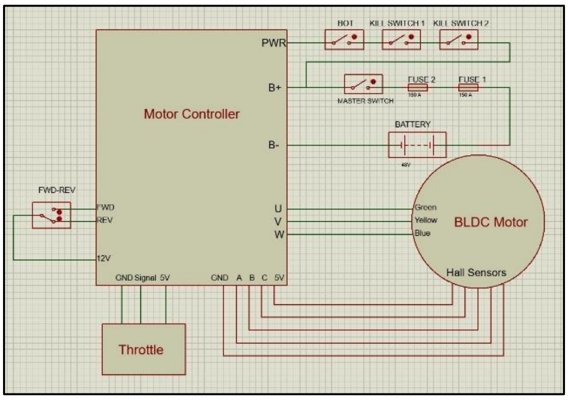


Fig. No. 1.1.33 Wiring diagram of motor controller

BATTERY DESIGN REPORT: -

Tractive System Accumulator: -

➢ **Type:** Lithium Ferro Phosphate Battery (LiFePO4)

➢ **Dimensions of battery pack and cell: -**

No. of battery pack = 01

The cells used in the battery pack are 16 prismatic cells connected in series having,
Length = 220 mm
Width = 130.2mm
Thickness 40.2 mm

Dimensions of Battery Pack:
Length = 362mm
Width = 445mm
Height = 300mm

Battery specifications:

1.	Nominal voltage	51.2 V
2.	Charge voltage	58.4 V
3.	Capacity	100 Ah
4.	C-rate (charging)	0.5C
5.	C-rate (discharging)	3C
6.	IP rating	IP67
7.	Depth of Discharge	85%
8.	SOH	99.67%
9.	Discharge Cutoff Voltage	40V
10.	Charging Cutoff Voltage	58.4V
11.	Discharge Cutoff Voltage of cell	2.5V
12.	Charging Cutoff Voltage cell	3.65V
13.	Life Cycle of Cell	≥5000 cycles

Table No. 1.1.16 Specifications of Battery

Battery Management System:
Centralized BMS from JBD is used.

No. of BMS	1
Cell voltage range	2.2 V - 3.75 V
Continuous charging and discharging current (Range)	40A -200A
Cell strings range	14-24S
Over-current cutoff (discharge)	150A

Table No. 1.1.17 Specifications of BMS

SOC Calculations:
State of Charge (SOC) represents the available capacity of the battery expressed as a percentage of its nominal capacity. We used Coulomb Counting method to estimate the SOC.

SOC = SOC initial - $\frac{I*t}{c}$ *100

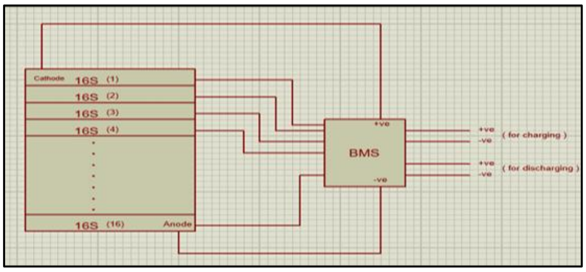


Fig. No. 1.1.34 battery pack connection diagram

SLI BATTERY:
The Lead-acid auxiliary battery is used**.**

Battery Type	Lead-acid
Battery Voltage	12V
Battery Capacity	10Ah

Table No. 1.1.18 specifications of SLI battery

Selection criteria for battery
For traction battery:

1) **Safety:** they are inherently safer due to their stable chemical composition and less prone to overheating and catching fire.
2) **Life cycle:** It offers a longer cycle life, crucial for long- term performance and endurance in electric vehicles
3) **Thermal stability:** they can operate reliably in a wider temperature range. This is particularly beneficial for applications in extreme environments.
4) Prismatic cells are larger than cylindrical cells, their higher volumetric energy density and simpler packaging can make them more space-efficient in certain battery

applications.

5) The motor is 48V and 3000W and according to calculations of current requirement for torque we have selected 51.2V battery with capacity of 100Ah.

For SLI battery:

1) It doesn't require any maintenance.
2) It is best in terms of reliability and working capabilities.
3) It offers the longest life
4) 97% of Lead can be recycled and reused.
5) All the low voltage components work on 12VDC and require 3A or less that is why we have chosen he battery with 12V and 10Ah capacity.

Calculations of charging and discharging battery packs (tractive and SLI)

For Tractive Accumulator: -
Charging rate:
The charger's output is 58.4 V, 20A.
As we get 20A the battery is completely charged from 0% to 100 % in,
=100Ah/20A = 5hours.

For SLI battery:
Charging rate:
The charger output is 15V, 2A So, battery is completely charged in,
= 10Ah / 2A = 5 hours

For Tractive Accumulator: -
Discharging rate:

Rated power of motor = 3000W

Battery capacity = 5120Wh.

Total Discharge time = 5120/3000 = 1.7hr.

For SLI battery:
Discharging rate:

Total wattage required by low voltage components = ~50W.
Battery capacity = 120Wh
Total Discharge time = 2.4Hrs

Calculations to justify the range of the vehicle with respect to capacity.

If we are travelling at a speed of 60Kmph

Range of vehicle = 60*1.7 = 102Km

DC Converter:

We are going to use DC-DC converter to power the Autonomous system in our vehicle.

Specifications: -

Parameter	Value
Input Voltage	24-72V
Output Voltage	12V
Current Rating	10A

Table NO.1.1.19 DC Converter Specifications

Safety Components:-

HRC Fuse :

Fig No. 1.1.35 HRC Fuse

An HRC fuse is used in a solar electric vehicle DC system to protect the battery and power electronics from overcurrent and short circuits. When excessive DC current flows, the fuse element melts and the quartz sand inside quenches the arc, safely interrupting the circuit and preventing damage.

Parameter	Value
Voltage rating	500VAC
Current rating	160A
Rupturing Capacity	>80kA

Table NO.1.1.20 DC Fuse Specifications

MCCB: -

A DC MCCB (Molded Case Circuit Breaker) is used in a solar electric vehicle to protect the DC system from overcurrent and short-circuit faults. When a fault occurs, the MCCB automatically trips using thermal and magnetic mechanisms, and its arc-chute system safely extinguishes DC arcs, allowing reliable isolation and reset of the circuit.

Fig. No. 1.1.36 DC MCCB

Parameter	Value
Voltage rating	220VDC
Current rating	160A
Ultimate breaking Capacity	10kA

Table NO.1.1.21 DC MCCB Specifications

SSR: -

A Solid-State Relay (SSR) is used in a solar electric vehicle DC system for fast and reliable switching of electrical loads. It uses semiconductor devices instead of mechanical contacts, enabling silent operation, high switching speed, and electrical isolation between control and power circuits.

Fig. No. 1.1.37 SSR

Parameter	Value
Input Voltage	3-32VDC
Current rating	60A
Output	4-60VDC

Table NO.1.1.22 SSR Specifications

SOLAR DESIGN REPORT

Panel Specifications:

Type	Monocrystalline
Number of panels	8
Voc	70.04V
Vmpp	61.8V
Isc	12.67A
Impp	11.6A
Power at STC condition	716.88W
Efficiency	15.81%

Table No. 1.1.23 Panel Specifications

Solar cell specifications:

Voc	0.68V
Vmpp	0.6V
Isc	6.38A
Impp	5.8A
Power at STC Condition	3.48W
No. of Cells	206

Table No. 1.1.24 Solar cell specifications

Basis for Selection of Solar Panel Specifications and Configuration:

- Available Vehicle Surface Area
  The usable top surface area of 3.4 m^2 limited the maximum number of solar cells and total panel size.
  $P_{max}=3.4\times1000\times0.1581=537.54$ W

- Solar Cell Selection and Efficiency
  High-efficiency monocrystalline solar cells (15.81%) were selected to maximize power generation within limited area.

- Battery Voltage Requirement (48 V System)
  The panel operating voltage was designed higher than battery voltage to enable efficient MPPT operation.
  $V_{mp,string}=61.8\ V>48\ V$
- MPPT Controller Compatibility Check
  $V_{oc,max}=87.55\ V<100\ V$
  $I_{max}=15.95\ A<20\ A$

- Structural and Weight Considerations
  Lightweight panel construction compatible with polycarbonate bodywork was selected to minimize vehicle mass.

Calculation of Solar irradiance and Solar irradiation:

Solar irradiance (STC) : 1000W/m^2
Total Panel area (A) = 3.4m^2
Solar irradiation
For 1 hr = 1kwh/m^2
For 5 peak sun hrs: 5kwh/m^2

Energy generated by panel = p* t
716.88*5 = 3.58 kwh/day

Calculation of running the vehicle with solar panel alone:

Panel Power = 716.88 W
Power ratio = Psolar/Pmotor
= 716.88/3000
= 0.2398*100%
= 24%
At running torque i.e., 22.95Nm speed is 63.83kmph
Speed formula = Speed at nominal condition* ratio
At = 63.83kmph*0.239
=14.3kmph

Therefore, speed of vehicle when running only on solar is achieved 14.3kmph

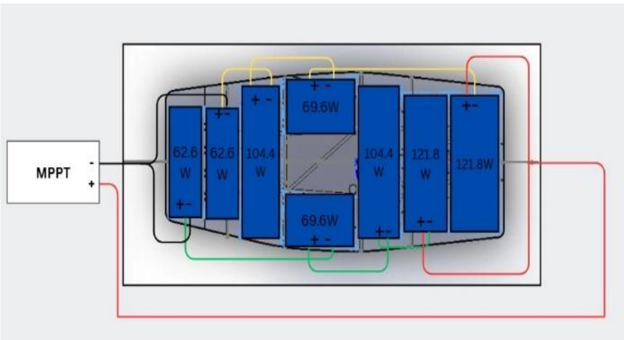


Fig No.1.1.38 Panel Connection Diagram.

Solar Charge Controller:

Type	MPPT
Input Voltage	100V DC
Output Voltage	60V
Maximum Output Current	20A
Load Voltage	48V
Type of Algorithm Used	P&O (Perturb & Observe)

Table No.1.1.25 Solar Charge Controller Specifications

Selection criteria for Solar charge controller: -

- Supports 48 V battery system, matching the vehicle electrical architecture.
- 100 V maximum PV input rating safely accommodates the solar panel open-circuit voltage.
- 20 A output current capacity is sufficient to handle the system's maximum solar power.
- MPPT technology ensures maximum power extraction under varying irradiance and temperature conditions.
- High conversion efficiency (>98%) reduces energy losses.
- Integrated protection features ensure safe and reliable operation.
- Proven reliability and build quality suitable for mobile and automotive applications.

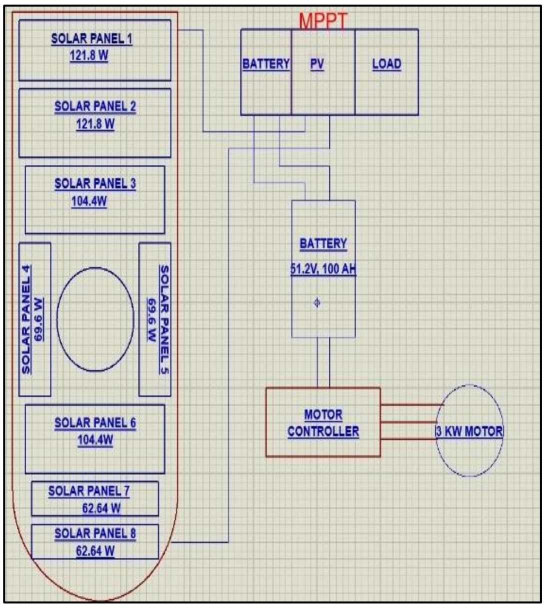


Fig No.1.1.39 Basic level circuit of charge Controller

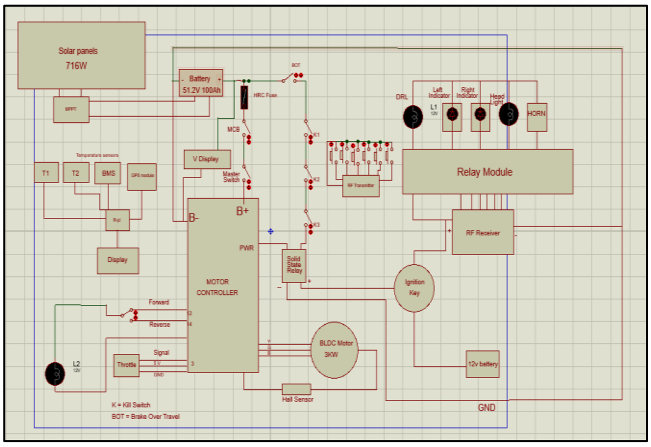


Fig No.1.1.40 Vehicle Circuit Diagram

VEHICLE CAD MODEL

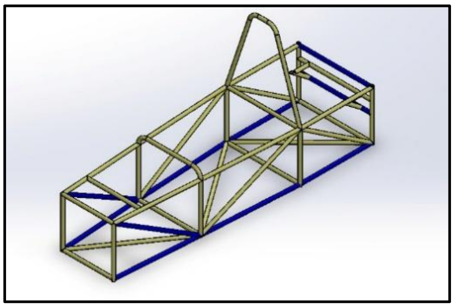


Fig. No. 1.2.1 Isometric view

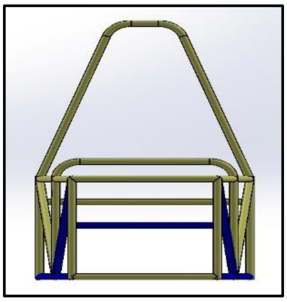


Fig. No. 1.2.2 Front view

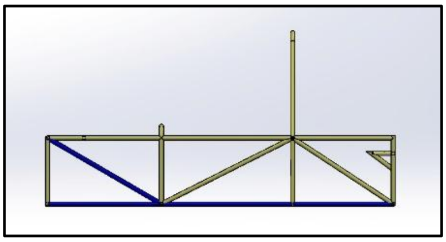


Fig. No. 1.2.3 Side view

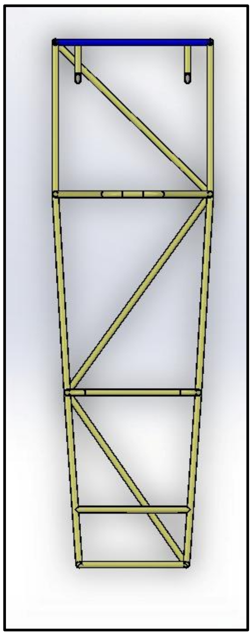


Fig. No. 1.2.4 Top view

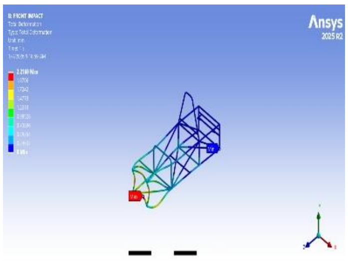
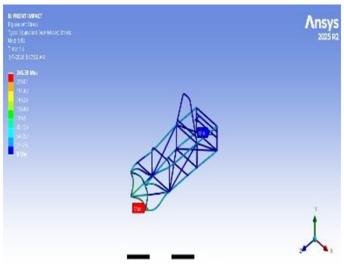
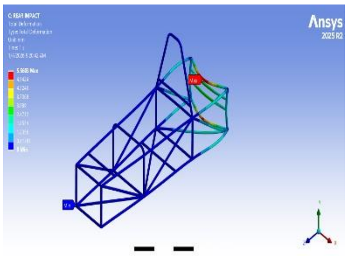
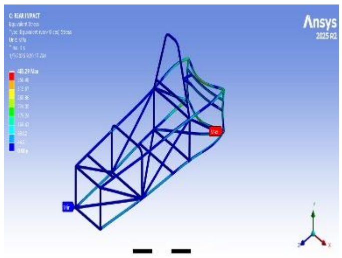
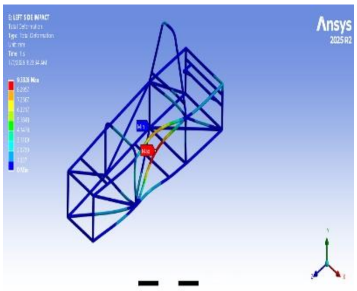
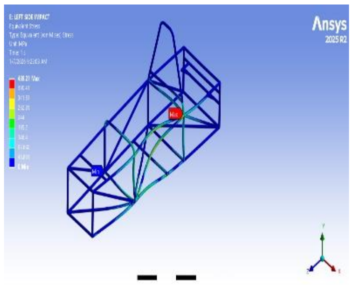
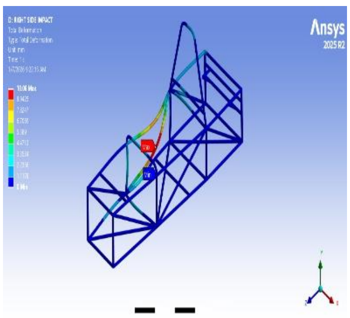
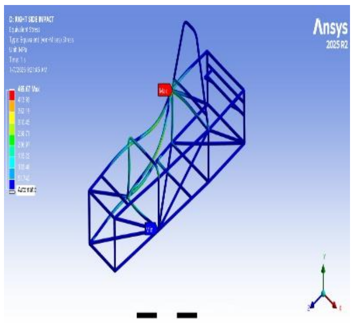
Structural Assessment

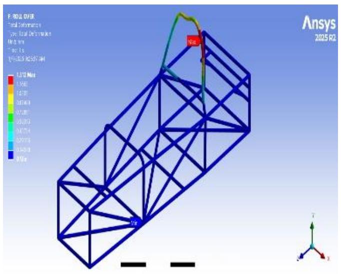
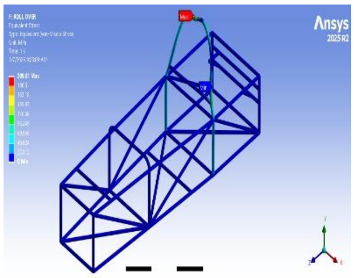
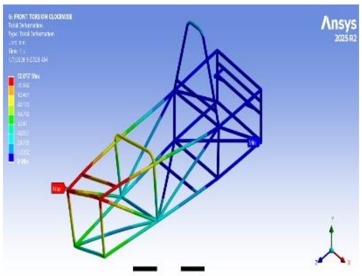
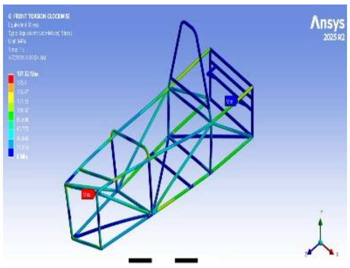
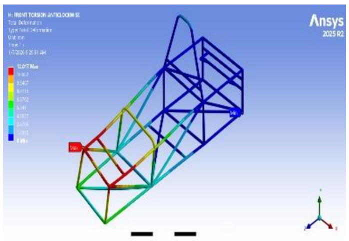
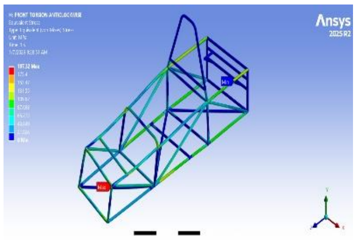
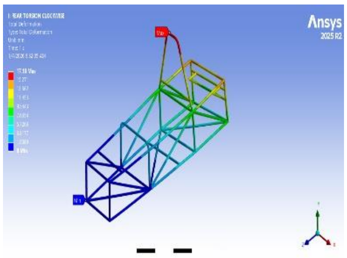
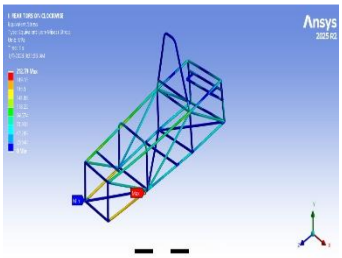
- Roll Cage Analysis: The design and development process of the roll cage involve various factors; namely material selection, frame design, cross-section determination and finite element analysis. One of the key design decisions of our frame that greatly increases the safety, reliability and performance in any automobile design is material selection. To ensure that the optimal material is chosen, extensive research was carried out and compared with materials from multiple categories. Roll Cage was designed in SOLIDWORKS 2025 & Analyzed in ANSYS 2025 R2. Static Analysis, Modal analysis and Explicit Dynamics was performed.

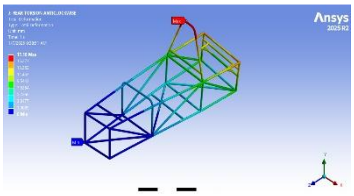
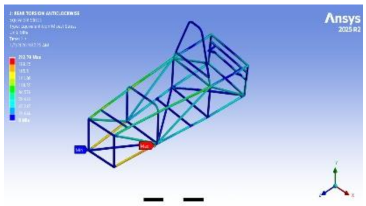
TYPE OF ELEMENT

Meshing: Meshing is the process of dividing a complex geometry into smaller, finite elements for numerical analysis, commonly used in Finite Element Analysis (FEA). The accuracy, efficiency, and computational cost of the analysis depend heavily on meshing quality. We have used 1D meshing.

Technical properties	AISI 4130 (25.4-2*1.6 mm)	AISI 4130 (25.4-2*1.2 mm)
Density	7.85 g/cm^3	7.85 g/cm^3
Young's modulus	205GPa	205Gpa
Diameter	OD 25.4 mm ID 22.2 mm	OD 25.4 mm ID 22.9 mm
Thickness	1.6mm	1.25mm
Tensile strength	760.91 MPa	737.24 MPa
Ultimate yield strength	804.45 MPa	783.39 MPa

Types of Analysis	Load Calculated	Deformation (Maximum)	Stress Induced (Maximum)	FOS
Frontal Impact	Mass of vehicle = 283.02kg Initial velocity of vehicle(vi)= 0 m/s Final velocity of vehicle(of) = 16.67 m/s Impact time (t) = 0.25s Formula used, ΔP = Ft move - m.vi = F t 283.02 x 16.67 – 283.02 x 0 = F x0.25 F = 18870 N	 Deformation – 2.21 mm	 Stress Induced – 246.39 Map	2.99
Rear Impact	Mass of vehicle = 283.02kg Initial velocity of vehicle(vi)= 0m/s Final velocity of vehicle(vf) = 16.67 m/s Impact time (t) = 0.25s Formula used, ΔP = Ft m.vf - m.vi = F t 283.02 x 16.67 – 283.02 x 0 = F x 0.25 F = 18870 N	 Deformation – 5.56 mm	 Stress Induced – 403.29 Mpa	1.82
Left side impact	Mass of vehicle = 283.02kg Initial velocity of vehicle(vi)= 0m/s Final velocity of vehicle(vf) = 16.67 m/s Impact time (t) = 0.25s Formula used, ΔP = Ft m.vf - m.vi = F t 283.02 x 16.67 – 283.02 x 0 = F x 0.25 F = 18870 N	 Deformation – 9.33 mm	 Stress Induced – 439.21 Mpa	1.82
Right side impact	Mass of vehicle = 283.02kg Initial velocity of vehicle(vi)= 0m/s Final velocity of vehicle(vf) = 16.67 m/s Impact time (t) = 0.25s Formula used, ΔP = Ft m.vf - m.vi = F t 283.02 x 16.67 – 283.02 x 0 = F x 0.25 F = 18870 N	 Deformation – 10.06 mm	 Stress Induced – 465.67 Mpa	1.58

Roll-Over Impact	Impact time t= 0.25s During the fall, the whole potential energy changes into kinetic energy, $mgs = mv^2/2$, $v = \sqrt{(2gh)}$ $v=\sqrt{(2\times9.81\times3.048)}$ v = 7.73m/s Now, By Impulse Momentum Theory, the force F can be calculated as F= mv/t = (283.02 ×7.73)/0.25 F= 8790.57 N	 Deformation – 1.312 mm	 Stress Induced – 209.81 Mpa	3.51
Front Torsional Clockwise	Calculation for torsional Force Mass of vehicle (m) = 283.02 Kg Force (F) = 1.4G = 1.4×9.81×283.02 F = 3886.99 N	 Deformation – 12.01 mm	 Stress Induced –197.32 Mpa	3.73
Front Torsional Anticlockwise	Calculation for torsional Force Mass of vehicle (m) = 283.02 Kg Force (F) = 1.4G = 1×9.81×283.02 F= 3886.99 N	 Deformation – 12.01 mm	 Stress Induced –197.32 Mpa	3.73
Rear Torsional Clockwise	Calculation for torsional Force Mass of vehicle (m) = 283.02 Kg Force (F) = 1.4G = 1.4×9.81×283.02 F = 3886.99 N	 Deformation – 17.18 mm	 Stress Induced – 212.79 Mpa	3.46

Rear Torsional Anticlock wise	Calculation for torsional Force Mass of vehicle (m) = 283.02 Kg Force (F) = 1.4G = 1.4×9.81×283.02 F = 3886.99 N	 Deformation – 17.18 mm	 Stress Induced – 212.79 Mpa	3.46

MODAL ANALYSIS

In chassis design, the designer considers not only the vehicle's bending, torsion, braking, and turning conditions but also the natural frequency of the chassis and the excitation frequency of the gearbox. We use 1000 Hz as the excitation frequency. The maximum number of modes chosen for analysis is 10.

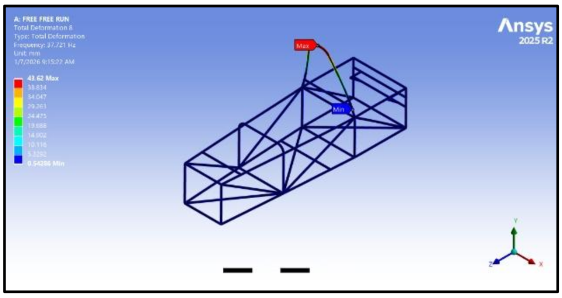


Fig.No. 1.2.5. Total Deformation

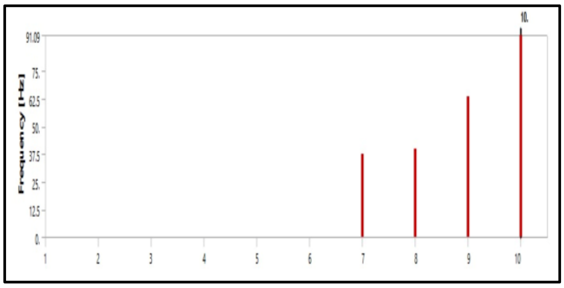


Fig. No. 1.2.6. Frequency Graph

Explicit Dynamics

In our vehicle, we implemented Explicit Dynamics in ANSYS to accurately simulate high-speed, time-dependent events such as frontal impacts, side collisions, and rollover scenarios. Unlike standard static analysis, Explicit Dynamics calculates the response of the chassis and structural members for rapidly changing forces over very small-time intervals, capturing non-linear behavior, plastic deformation, buckling, and large displacements that occur during real-life crashes. By applying realistic boundary conditions and material properties, including AISI 4130 Chromoly steel with its yield strength and stiffness, we were able to predict the maximum equivalent stress, combined stress, and deformation of each member. This allowed us to identify potential failure points, evaluate the factor of safety for all structural components, and optimize the geometry, thickness, and material distribution to enhance vehicle safety while maintaining lightweight construction. Overall, the use of Explicit Dynamics enabled precise pre-testing predictions, ensuring that the vehicle structure could withstand dynamic loads without physical prototyping, thereby saving time and resources while improving reliability and safety.

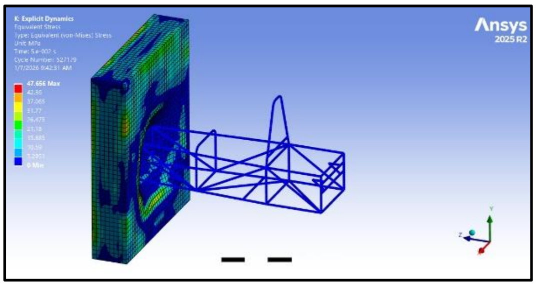


Fig. No. 1.2.7. Equivalent Stress

Bending strength & bending stiffness

- **Bending strength & bending stiffness (1.6 mm)**

Bending strength = M*y/ I
I = 8521.96 mm4
Z = I / y = 671.02 mm^3
Bending strength = 509793.9*12.7/8521.96
Bending strength =760.90 MPa
Bending stiffness BS = E x I = 205x103 x 8521.96 = 1.78 x10^9 N/mm^2

- **Bending strength & bending stiffness (1.2mm)**

Bending strength = M*y/ I
I = 6694.93 mm^4
Z = I / y = 527.57 mm^3

Bending strength = 388782.5*12.7/6.694.93
Bending strength = 737.27 MPa
Bending stiffness BS = E x I = 205x103 x 6694.03 = 1.40×10^9 N/mm^2

SR No	Thickness of pipe	Bending strength	Bending stiffness
1	1.6 mm	760.90 Mpa	$1.78*10^9$
2	1.2 mm	737.27 Mpa	$1.40*10^9$

Table No. 1.2.1. Bending Strength & Stiffness

INNOVATION

Implementation of Touch Sensor Based Intelligent Speed Control System

1. Problem Statement

Drivers often lose attention or remove hands from the steering wheel, especially during long drives or stressful situations. This can lead to accidents or loss of vehicle control. Current vehicles lack a simple, low-cost system to monitor hand placement and alert drivers automatically.

2. Scope of Innovation

- Enhances driver safety by ensuring hands remain on the steering wheel.
- It can be integrated into cars, buses, trucks, and driving schools.
- Encourages safe driving habits.
- Provides automatic intervention in case of negligence.

3. Existing Idea

- Many vehicles have alarms for seat belts or lane departure.
- Some high-end cars use advanced driver monitoring (cameras, eye-tracking) to detect fatigue.
- These systems are either expensive or complex to integrate into small vehicles.

4. Proposed Idea

- Our innovation is a Hands-On Steering System
- Touch sensor on the steering wheel to detect hand placement.
- Buzzer alert if hands are absent.
- Automatic reduction of vehicle power if the alert is ignored.
- A control unit that processes sensor input and triggers safety interventions automatically.

5. Justification of Novelty

- Simple and cost-effective compared to camera-based systems.
- Focuses on hands-on safety, not just eye fatigue.
- Can be implemented in any vehicle, including student-built or commercial vehicles.
- Fully automatic intervention enhances safety without relying on driver reaction.

6. Product Value Added to the Vehicle

- Improves driver safety and reduces accident risk.
- Encourages responsible driving behavior.
- Provides a low-cost solution for both personal and fleet vehicles.
- It can be easily retrofitted in existing vehicles.

7. Innovation Budget

✓ Component Approx. Cost (INR)

- Touch Sensor - 300
- Buzzer - 100
- Microcontroller / Control Unit - 500
- Power Control Circuit - 400
- Miscellaneous (wires, connectors, etc.) - 200
- Total Estimated Cost - ₹1500

8. Feasibility

- Uses off-the-shelf sensors and components.
- Easy to install and integrate in a vehicle.
- Low maintenance and does not interfere with normal driving.

- Scalable for multiple vehicle types.

9. Image of the Innovation

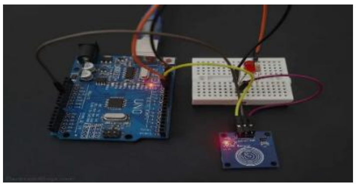


Fig No. 1.3.1 Innovation

Implementation of LiDAR-Based Intelligent Speed Control System

1. Problem Statement

Vehicles often face sudden bumps, potholes, or uneven surfaces on roads, leading to passenger discomfort, damage to vehicle suspension, and potential accidents. Manual speed control by drivers may not be efficient in detecting such irregularities, especially at high speed or during low visibility conditions.

2. Scope of Innovation

- Real-time detection of road irregularities.
- Automatic speed adjustment to enhance safety and comfort.
- Applicable to all vehicles: cars, buses, and commercial vehicles.
- Reduces maintenance cost due to fewer suspension damages.
- Can integrate with future autonomous vehicle systems.

3. Existing Idea

Current systems include:

- Mechanical sensors or bump sensors: Detect only when contact occurs.
- Camera-based detection: Depends on light conditions and image processing speed.
- Driver-based control: Relies on human reaction time, which may be delayed.
- Limitation: Existing systems are either reactive (after hitting the bump) or highly environment dependent.

4. Proposed Idea

- Use LiDAR (Laser Detection and Ranging) to scan the road surface continuously.
- Measure road height deviations in real-time before the vehicle reaches them.
- Microcontroller processes the data and calculates required speed reduction.
- Automatically controls throttle or braking system to adjust speed safely.

5. Justification of Novelty (Explanation and Calculations)

Novelty:

Unlike existing mechanical or camera-based methods, LiDAR is non-contact, real-time, and accurate, detecting bumps before the vehicle reaches them.

Calculations Example (Placeholder):

- Vehicle speed: v = 40 km/h 11.11 m/s
- Detection distance of LiDAR: d = 5 m
- Reaction time: $tr = d/v$=5/11.11 ≈ 0.45 Seconds
- Required deceleration to safely cross bump height h = 0.1m
- h=0.1m:
- $a = v^2/2d$= $(11.11)^2$ / 2.5 ≈ 12.34 m/s^2

This shows that the vehicle can safely reduce speed before hitting the bump, validating the system's effectiveness.

6. Product Value Added to the Vehicle

- Ensures passenger comfort.
- Reduces suspension and vehicle damage.
- Enhance road safety and reduces accident risks.
- Adds intelligent automation for modern vehicle systems.
- Can be integrated with autonomous driving modules.

7. Budget of the Innovation (Approximate)

Component Approx. Cost (₹)

- LiDAR Sensor = 15,000
- Microcontroller Unit = 2,000
- Motor/Throttle Controller = 3,000
- Mounting & Wiring = 2,000
- Miscellaneous = 1,000
- Total = 23,000

8. Feasibility

Technical Feasibility: Uses widely available LiDAR sensors and microcontrollers.

- Economic Feasibility: Moderate cost; high value in safety and reduced maintenance.
- Operational Feasibility: Easy to implement in modern vehicles without major modifications.
- Scalability: Can be applied to fleets, commercial vehicles, and autonomous cars.

Autonomous Driving Report

1. Introduction

Self-driving vehicles use sensors, cameras, and AI to navigate without human input. These technologies enable environmental perception, mapping, obstacle identification, and automated driving decisions (steering, acceleration, braking).

2. Motivation

Autonomous vehicle development is driven by the need for safer roads (reducing human error-caused accidents), improved efficiency (traffic flow, fuel use), enhanced accessibility (for those unable to drive), reduced congestion, increased productivity, economic benefits, and a more convenient travel experience.

3. Levels of Autonomy

The Society of Automotive Engineers have defined levels for self-driving vehicles are as follows:

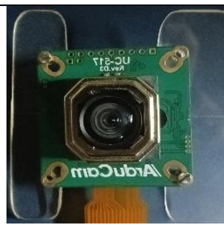
LEVEL 0: No autonomy, driver has full control over the vehicle.

LEVEL 1: The basic driver assistance is provided to the driver (Cruise Control, ABS, etc.)

LEVEL 2: The vehicle steers, accelerates, applies brakes when necessary on its own in monitoring of the driver.

Fig 1.4.1 Levels of Automation

4. Components

1.	Raspberry pi 4 b
	
	Raspberry Pi 4 Model B is a compact, powerful single-board computer designed for a variety of applications. Features a quad-core ARM Cortex-A72 processor running at 1.5 GHz, with RAM options of 2GB, 4GB, or 8GB LPDDR4, enabling smooth multitasking and performance with Gigabit Ethernet, Wi-Fi (802.11ac), and Bluetooth 5.0 for versatile networking options.
2.	ArduCam IMX477 Module
	
	It features a 12.3-megapixel Sony IMX477 sensor, offering high resolution and excellent low-light performance.
3.	Electric Linear Actuator

	
	Linear actuator - a machine which works on the mechanism of converting rotational motion into a push or pull linear motion. Commonly used in robotics, automation and industrial applications. Input: 12V DC Load capacity: 1500N Stroke length: 100mm Speed: 15mm/sec Frequency: 20%
4.	Arduino UNO
	
	The Arduino Uno is a versatile microcontroller board based on the ATmega328P, ideal for electronics projects and prototyping with its digital/analog I/O capabilities and user-friendly programming interface.
5.	H-Bridge Motor Driver L298N
	
	The L298N H-Bridge motor driver allows for the control of DC motors and stepper motors, enabling bidirectional rotation and speed control via PWM signals, making it ideal for robotics and automation projects.

6.	DC 12V Relay Receiver Module RF Transmitter 433Mhz Wireless Remote-Control Switch
	
	DC 12V 1CH Relay Receiver Module RF Transmitter 433Mhz Wireless Remote-Control Switch

5. Autonomous Circuit Diagram

Autonomous System Architecture

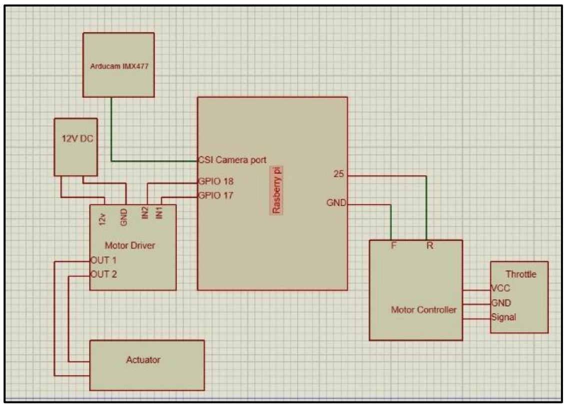


Fig 1.4.2 Autonomous Circuit Diagram

6. Autonomous Core Implementations

6.1 Software Terminologies

Python, Computer Vision, YOLO machine learning

6.2 Implementation

Lane detection works by using a camera to continuously capture the road ahead and processing each frame to identify the left and right lane markings. From these detected lanes, the center of the lane is calculated and compared with the center of the camera view to find the error. This error represents how much the vehicle is drifting from the lane center, and a PID controller uses it to smoothly adjust the steering. By repeating this process for every frame, the vehicle stays centered within the lane while driving.

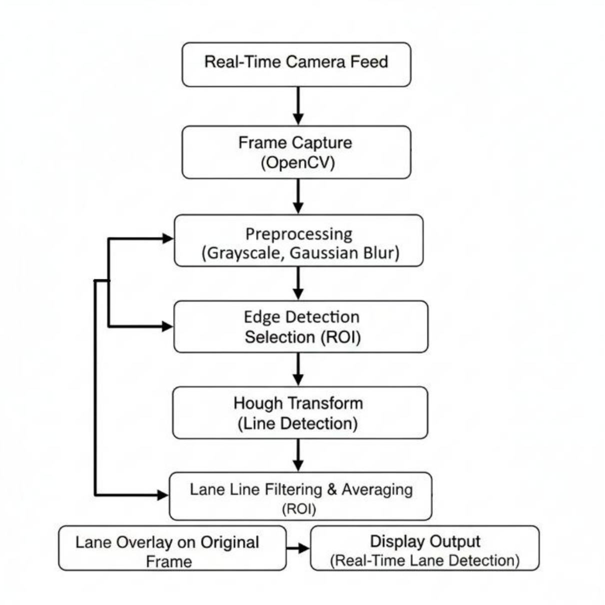


Fig. 1.4.3: Lane Detection

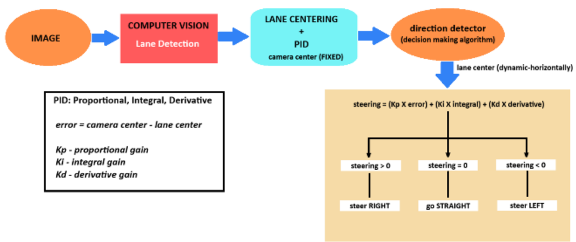


Fig. 1.4.4: Autonomous Decision Flow

7. Automotive Safety and Technologies

ADAS (Advanced Driver Assistance Systems) belongs to the immersive field of automotive safety and automotive technologies. Features such as Cone detection, path maintaining. Several disciplines are included:

1. Automotive Engineering: Design and development of the vehicle systems that enhances safety, comfort and performance.
2. Computer Vision & Machine Learning: Real-time detection of the objects, lanes, vehicles, pedestrians within the environment of the vehicle, enabling the features like automatic braking, lane-keeping and traffic sign recognition.
3. Sensor Technology: ADAS relies on various sensors such as cameras, LiDAR, Radars, ultrasonic sensors to gather the information about the vehicle surroundings.
4. Control Systems: Includes the algorithms and systems for processing sensor data and taking appropriate actions, such as braking and steering.
5. Human-Machine Interface (HMI): Interface which provides feedbacks and also alerts to driver about system's status or any potential hazards.

According to Society of Automotive Engineers (SAE), ADAS fits within the vehicle automation spectrum with many of its features classified as Level 1 or Level 2 automation.

8. Emergency Safety in Vehicle

A **433MHz RF Wireless Remote-Control Switch** with a **12V DC Relay Receiver Module** is used for **emergency stopping** of the vehicle. Pressing **'open'** on the remote **turns off** the **Solid-State Relay (SSR)**, cutting the tractive supply, while **'close'** turns it **on**, restoring power.

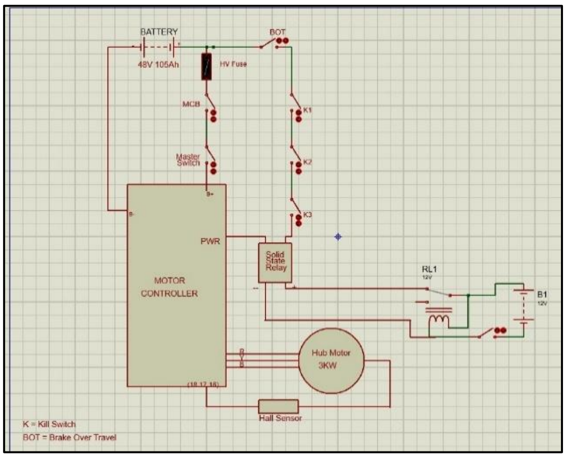


Fig. No. 1.4.5 Emergency Safety Circuit

9. Integration of automation technology in vehicles

9.1 Steering (Rack and Pinion System)

The integration of electric linear actuator in rack and pinion steering system represents a significant advancement in automotive technology. This approach enhances steering precision, reduces mechanical complexity, and supports the development of advanced driver assistance systems (ADAS) and fully autonomous vehicles. As the automotive industry continues to evolve, the role of servo motors in steering technology will become increasingly crucial in improving safety, performance, and the overall driving experience.

Fig No.1.4.6 Rack and Pinion System Linear Actuator Actuation

9.2 Braking Actuation

The use of a linear actuator for automated brake pedal control represents a crucial innovation in vehicle automation. By enabling real-time adjustments to braking based on obstacle detection, this technology enhances both safety and driving comfort. As vehicles evolve towards greater automation, such systems will play an essential role in improving overall performance and safety.

The integration of linear actuators in automotive systems represents a significant advancement in vehicle automation and safety.

Fig 1.4.7 Brake Pedal Control with Actuator

10. Alterable (manual to autonomous mode and vice versa):

To changeover from manual to autonomous level, implementation of a toggle switch for easy to access to the driver is implemented. So, the driver can have full access with the changeover. The toggle switch's selected toggle state will signify the changeover either from manual or to autonomous mode.